

Equipment Maintenance Tracker

Application Name: NexOps v3.0 - Industrial Equipment Maintenance Tracker

Technology Used: Python, Flask, HTML, CSS, JavaScript, Chart.js

Data Storage: CSV File (maintenance_records.csv)

Deployment: Local (localhost:5000) + PythonAnywhere Cloud

Section 1: UI Design

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NEXOPS v3.0

Marcus Miller

MM

Dashboard

Log Issue

Reports

Equipment List

Settings

Home / Log Issue

ISSUE LOGGING STATION

Monday, June 15, 2026 at 01:36:12 AM

Technician portal to log active equipment failures and safety hazards.

EQUIPMENT ID

e.g. EQ-832

EQUIPMENT NAME

e.g. Main Intake Valve B

EQUIPMENT TYPE

Select equipment type

SEVERITY LEVEL

Select severity

PRIORITY LEVEL

Select priority

LOCATION / ZONE

Select location / zone

ESTIMATED REPAIR TIME

Select estimated repair time

INITIAL STATUS

Open

TECHNICIAN NAME

e.g. Marcus Miller

DATE IDENTIFIED

06/15/2026

ISSUE DESCRIPTION

Provide a detailed description of the fault, failure codes, and any immediate safety actions taken...

SUBMIT RECORD

Factory Node: TX-H1

Server Time: 01:36:12

Section 2: Data Source

	A	B	C	D	E	F	G	H	I	J	K
1	equipment_id	equipment_name	equipment_type	issue_description	severity	technician_name	date	status	priority	location	est_repair_time
2	EQ-PL-001	ASML Twinscan Align B	Photolithography Machine	Critical alignment failure on lens module during calibration phase. Required field service engineer dispatch.	Critical	Alex Carter	6/5/2026	Open	P1-Critical	Clean Room A	More than 2 days
3	EQ-CVD-002	Applied Materials CVD 12	Chemical Vapor Deposition (CVD) Chamber	Mass flow controller sensor mismatch causing deposition thickness deviations. MFC replaced.	High	Sarah Jenkins	6/7/2026	In Progress	P2-High	Clean Room B	4-8 hours
4	EQ-WIS-003	KLA Tencor Wafer Inspect	Wafer Inspection System	Camera calibration drift on laser lens unit. Realigned and recalibrated.	Medium	Marcus Miller	6/8/2026	Resolved	P3-Medium	Clean Room A	1-4 hours
5	EQ-ROB-004	Fanuc Cleanroom Arm	Industrial Robot Arm	Overcurrent fault on joint3 motor. Cleared debris and checked cabling.	High	John Doe	6/10/2026	In Progress	P2-High	Fab Floor 2	1-4 hours
6	EQ-CNC-005	Haas VF-2 Spindle	CNC Milling Machine	Spindle vibration exceeded warning threshold. Tightened mounts.	Low	Marcus Miller	6/11/2026	Resolved	P4-Low	Maintenance Bay	Less than 1 hour
7	EQ-CT-006	Marley Cooling Tower B	Cooling Tower	Cooling water pump motor failed to start due to contactor damage. Replacement ordered.	Critical	Alex Carter	6/2/2026	Open	P1-Critical	Utility Room	1-2 days
8	EQ-XFMR-007	Substation Transformer 2	High Voltage Transformer	Transformer core temperature exceeded safe limit. Oil pump service initiated.	High	Sarah Jenkins	6/12/2026	In Progress	P2-High	Utility Room	4-8 hours
9	EQ-CAS-008	Atlas Copco Compressor	Compressed Air System	Pressure drop in cleanroom delivery line. Located and patched main line leaks.	Medium	John Doe	6/13/2026	Open	P3-Medium	Maintenance Bay	1-4 hours
10	EQ-ETC-010	Lam Research Etcher	Etching System	Chamber chemistry ratio deviation during silicon etch. Re-calibrated mass flow controllers.	High	Sarah Jenkins	6/15/2026	Open	P2-High	Clean Room B	4-8 hours
11	EQ-HVAC-009	Clean Room HVAC AHU-4	Clean Room HVAC	HEPA filter differential pressure high and filter change performed.	Low	Marcus Miller	6/14/2026	Resolved	P4-Low	Clean Room A	1-4 hours

Section 3: Structured Prompts

Prompt	Purpose	Description
Prompt 1	Project Structure & Input Form	Built full Flask project structure, professional input form with 11 fields, CSV data file, and dark industrial theme
Prompt 2	Data Visualization Charts	Added Chart.js bar chart for status distribution and donut chart for severity analysis reading live from CSV
Prompt 3	Search & Filter Features	Added real-time search bar and filter dropdowns for status and severity without page reload
Prompt 4	Export & Delete Features	Added Export CSV download button and Delete Record button with live KPI and chart updates
Prompt 5	Professional UI Redesign	Redesigned UI to look like big tech enterprise software with improved color scheme and layout
Prompt 6	Sidebar & Color Fixes	Fixed sidebar overlap bug, redesigned with Bloomberg/Palantir inspired dark professional theme
Prompt 7	New Pages & Full Test	Added working Reports, Equipment List, and Settings pages with Flask routes and full workflow test
Prompt 8	Navigation & Chart Fixes	Fixed navbar visibility, Home breadcrumb links, Equipment Type pie chart, ran full application test
Prompt 9	Light Mode & Complexity	Switched to professional white/light enterprise theme, added semiconductor equipment types and new form fields
Prompt 10	Logo & Final Bug Fixes	Fixed logo visibility, navbar contrast, home button routing, equipment breakdown chart display
Prompt 11	Seed Data & Chart Colors	Added 10 realistic semiconductor factory records, fixed pie chart with distinct colors per equipment type

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1. Prompt 1
Test
Please fix the following issues with the application:

1. MAINTENANCE PAGES
- The Reports, Equipment List, and Settings pages do not work when clicked. Create actual working pages for each.
- Reports page (reports.html): Show a summary table of maintenance statistics: total issues per equipment type, average resolution time, most common issue types. Include a line chart showing number of issues logged per day over time.
- Equipment List page (equipment_list.html): Show a comprehensive table of all current equipment: ID#, Name, Type, and Total Number of Issues Logged Against Them.
- Settings page (settings.html): Show a simple settings form with fields like company name, footer name, default technician name, and a toggle for email notifications. Does not need to save anything, just needs to look functional.
Add Flask routes in app.py for /reports, /equipment-list, and /settings.

2. COLOR AND DESIGN FIXES
- Keep the current color scheme completely, use a completely different approach inspired by Bloomberg terminal or Palantir Gotham - feels used by serious professionals
- Main background: pure near-black #000000
- Accent/Highlight: very dark charcoal #111111
- Borders: subtle medium gray suitable for structure
- Primary accent: vivid lime green #808000 ONLY for the most critical active elements like the active nav item indicator and primary action buttons
- Data colors: use a gradient set - cyan palette for informational, amber/orange for warnings, coral/red for critical, muted purple for resolved
- All numbers and IDs must use a monospace font like Courier New or Roboto Mono
- Bug fixes must use a clean sans-serif font
- Make it look like a serious professional tool not a website template

3. LAYOUT FIXES
- Remove the current generic logo box completely
- Replace it with a text based tag that says "Welcome" in a bold technical font
- Add a small version number tag next to it like "v2.1.4" in a muted color
- Add a thin horizontal separator line underneath the tag area in styles.css to separate it from the navigation links below

4. LAYOUT FIXES
- Keep the sidebar fixed on the left and make it 150px wide
- Make the main content area scrollable and responsive
- Implement a simple CSS scrollbar with height to collapse/expand the sidebar
- When the sidebar is collapsed, the main content area should expand to fill the screen
- Ensure all pages are responsive and look great on desktop, tablet, and mobile

5. DATA ENTRY VALIDATION & ONE MORE PAGE
- Add a custom validation check on the submit form to ensure all fields are filled, displaying a clean short notification telling the user which field is missing
- Fill in all fields with realistic test data and click Submit Record - verify the record saves correctly to maintenance_records.csv
- Verify the dashboard totals correctly after addition and the new record appears in the table
- Verify the site title updates with the correct counts
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2. Prompt 2

text

Please make the following major changes to the application:

1. RENAME THE APPLICATION:

- Change the app name from "Maintos" to "MexOps" - this should feel like a real semiconductor/industrial operations software product name
- update the version to v3.0
- update this name everywhere it appears in all HTML files, titles, and any text

2. COMPLETELY SWITCH TO LIGHT/WHITE MODE:

- Research and apply professional enterprise UX/UI design principles used by SAP, Oracle, Salesforce, and ServiceNow - these are the gold standard for industrial enterprise software
- Main background: clean white #ffffff
- Page/content background: very light gray #f4f6f9
- Card/panel background: pure white #ffffff with a subtle border #e0e0e0 and soft box shadow
- Sidebar background: clean white #ffffff with a left border accent
- Top navbar: pure white #ffffff with a bottom border #e0e0e0
- Primary accent color: deep professional blue #0055cc for buttons, active states, links
- Secondary colors: amber #f99e0b for warnings, red #dc2626 for critical, green #1e8449 for resolved, gray #607280 for neutral
- All text: dark charcoal #1a1a1a for headings, medium gray #405663 for body text, light gray #9ca3af for labels
- Table rows: white with alternating very light gray #f9f9f9, with blue left border on hover
- Remove ALL dark backgrounds completely

3. MAKE IT MORE COMPLEX AND REALISTIC FOR SEMICONDUCTOR/INDUSTRIAL USE:

- Research real equipment used at companies like TSMC, Intel, and Honeywell factories and update the Equipment Type dropdown to include real semiconductor and industrial equipment types such as:
 - * Photolithography Machine
 - * Chemical Vapor Deposition (CVD) Chamber
 - * Etching System
 - * Wafer Inspection System
 - * CNC Milling Machine
 - * Industrial Robot Arm
 - * Clean Room HVAC
 - * Conveyor Belt System
 - * Turbine Generator
 - * Cooling Tower
 - * High Voltage Transformer
 - * Compressed Air System
- Add a Priority Level field to the form (P1-Critical, P2-High, P3-Medium, P4-Low) separate from Severity
- Add a Location/Zone field to the form (dropdown: Clean Room A, Clean Room B, Fab Floor 1, Fab Floor 2, Utility Room, Maintenance Bay)

3. Prompt 3

text

Please fix the following bugs and issues found during testing:

1. NAVBAR VISIBILITY FIXES:

- The app name/logo "MexOps v3.0" is invisible because it blends into the white navbar background. Fix this by making the navbar background a deep professional dark navy #1e2a3a instead of white. Make the logo text white and bold so it stands out clearly.
- The user profile name "Marcus Miller" and avatar in the top right corner are also invisible. Make them white text on the dark navy navbar.
- Make all navbar icons and text white on the dark navy background.
- Keep the sidebar white as it is - only change the TOP navbar to dark navy.
- Add a clear visual separation between the dark top navbar and the white sidebar.

2. HOME BUTTON FIX:

- The home link in the breadcrumb navigation (Home / Dashboard) does not work when clicked. Fix it so clicking "Home" redirects to the dashboard page at route /dashboard.
- Make sure breadcrumb home links work correctly on ALL pages (Dashboard, Log Issue, Reports, Equipment List, Settings).

3. EQUIPMENT TYPE BREAKDOWN CHART FIX ON REPORTS PAGE:

- The Equipment Type pie chart on the Reports page is not showing any data. Debug and fix this.
- Make sure it reads data correctly from maintenance_records.csv and displays the breakdown by equipment type.
- If the chart library is not loading correctly, switch to using chart.js which is already working on the dashboard page.
- Test the chart with the existing records in the csv to make sure it displays correctly.

4. FULL APPLICATION WORKFLOW TEST:

- Open Log Issue form and fill in ALL fields with realistic semiconductor factory test data and submit. Verify it saves to CSV correctly.
- Go to Dashboard and verify the new record appears in the table and all KPI cards and charts update.
- Go to Reports page and verify ALL charts load with real data including the Equipment Type pie chart.
- Go to Equipment List and verify the new equipment appears.
- Test the search bar on the dashboard.
- Test the Status and Severity filter dropdowns.
- Test the Export CSV button.
- Test the Delete button on a record.
- Test clicking Home in breadcrumbs on every page.
- Fix every bug found during this testing before finishing.

Report exactly what was tested, what passed, and what was fixed.

4. Prompt 4

text

Please fix the following issues:

1. EQUIPMENT TYPE BREAKDOWN CHART COLORS AND DATA:

- The pie chart on the Reports page only shows 2 items and both are the same blue color. This looks empty and unprofessional.
- First, add sample/seed data to the maintenance_records.csv file with at least 8-10 realistic records covering many different equipment types so the charts look full and meaningful. Use these realistic semiconductor factory scenarios:
 - * A Photolithography Machine with a critical alignment failure
 - * A CVD Chamber with a gas flow irregularity
 - * A Wafer Inspection System with a camera calibration issue
 - * An Industrial Robot Arm with a joint motor fault
 - * A CNC Milling Machine with a spindle vibration warning
 - * A Cooling Tower with a water pump failure
 - * A High Voltage Transformer with an overheating alert
 - * A Compressed Air System with a pressure drop issue
 - * A Clean Room HVAC with a filter replacement needed
 - * An Etching System with a chemical concentration error
- Make each record have different technicians, dates spread across the past 2 weeks, different severity levels, priority levels, locations, statuses, and estimated repair times

2. FIX PIE CHART COLORS:

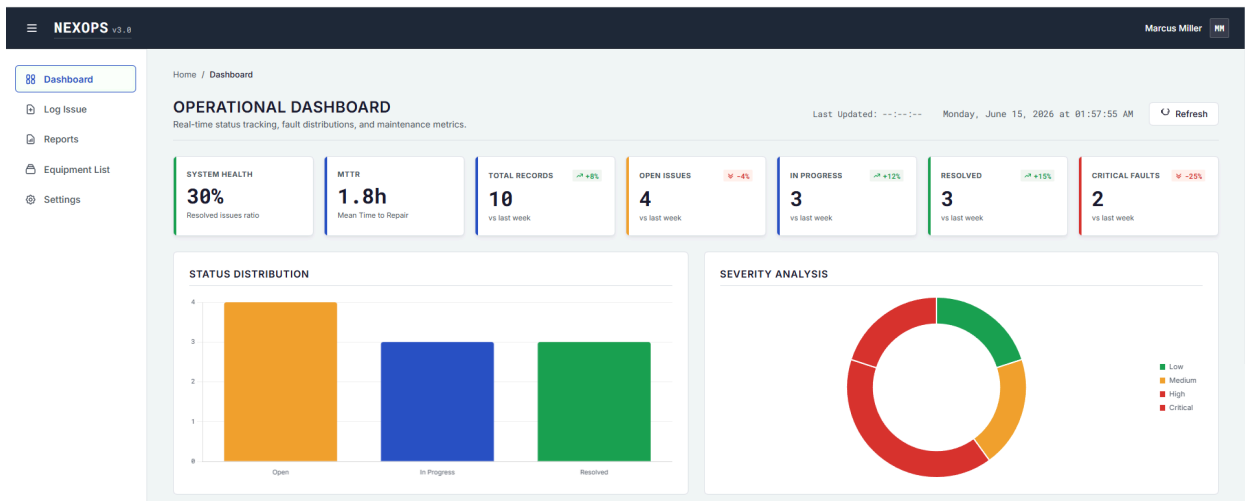
- Each slice of the Equipment Type pie chart must have a completely different distinct color. Use these specific colors for each equipment type:
 - * Photolithography Machine: #0055cc blue
 - * CVD Chamber: #7c3aed purple
 - * Wafer Inspection System: #0091b2 cyan
 - * Industrial Robot Arm: #005566 emerald
 - * CNC Milling Machine: #d97706 amber
 - * Cooling Tower: #dc2626 red
 - * High Voltage Transformer: #db2777 pink
 - * Compressed Air System: #6a3d9d lime
 - * Clean Room HVAC: #eas58bc orange
 - * Etching System: #db5cf6 violet
- Make the pie chart larger so it fills the container properly
- Add percentage labels on each slice

3. UPDATE ALL DASHBOARD CHARTS WITH NEW DATA:

- After adding the seed data, verify the Status Distribution bar chart and Severity Analysis donut chart on the dashboard also update and look full with varied data
- Make sure the KPI cards show realistic numbers based on the new seed data
- Make sure the Maintenance Activity logs table shows all records

Section 4: Visualized Output

Dashboard Charts:



Reports Charts:



Equipment List page:

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NEXOPS v3.0

Marcus MillerMM

Dashboard

Log Issue

Reports

Equipment List

Settings

Home / Equipment List

EQUIPMENT INVENTORY

Registered manufacturing hardware node records and associated logs counts.

Monday, June 15, 2026 at 02:00:18 AM

INVENTORY MASTER RECORDS

Q Filter equipment records (ID, name, type)...

EQUIPMENT ID	EQUIPMENT NAME	EQUIPMENT TYPE	TOTAL LOGGED ISSUES	CURRENT HEALTH STATUS
EQ-CAS-008	Atlas Copco Compressor	Compressed Air System	1	FAULT DETECTED (1)
EQ-CNC-005	Haas VF-2 Spindle	CNC Milling Machine	1	NOMINAL / CLEAR
EQ-CT-006	Marley Cooling Tower B	Cooling Tower	1	FAULT DETECTED (1)
EQ-CVD-002	Applied Materials CVD 12	Chemical Vapor Deposition (CVD) Chamber	1	FAULT DETECTED (1)
EQ-ETC-010	Lam Research Etcher	Etching System	1	FAULT DETECTED (1)
EQ-HVAC-009	Clean Room HVAC AHU-4	Clean Room HVAC	1	NOMINAL / CLEAR
EQ-PL-001	ASML Twinscan Align B	Photolithography Machine	1	FAULT DETECTED (1)
EQ-ROB-004	Fanuc Cleanroom Arm	Industrial Robot Arm	1	FAULT DETECTED (1)
EQ-WIS-003	KLA Tencor Wafer Inspect	Wafer Inspection System	1	NOMINAL / CLEAR
EQ-XFMR-007	Substation Transformer 2	High Voltage Transformer	1	FAULT DETECTED (1)

Factory Node: TX-H1

Server Time: 02:00:18

Settings page:

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NEXOPS v3.0

Marcus MillerMM

Dashboard

Log Issue

Reports

Equipment List

Settings

Home / Settings

SYSTEM SETTINGS

Configure the NexOps operational client properties and alerts parameters.

Monday, June 15, 2026 at 02:00:43 AM

COMPANY NAME

Apex Industrial Group

FACTORY NODE ID

TX-H1

DEFAULT TECHNICIAN NAME

Marcus Miller

SYSTEM MODE

Operational

UPTIME REFRESH INTERVAL

30 seconds

☒ Enable automated critical alert emails

SAVE CONFIGURATIONS

Factory Node: TX-H1

Server Time: 02:00:43

search and filter:

MAINTENANCE ACTIVITY LOGS									
Q HVAC					STATUS	All Statuses ▾	SEVERITY	All Severities ▾	EXPORT CSV
EQUIPMENT	TYPE	LOCATION/ZONE	PRIORITY	SEVERITY	ISSUE DESCRIPTION	TECHNICIAN	DATE	STATUS	ACTIONS
EQ-HVAC-009 Clean Room HVAC AHU-4	Clean Room HVAC	Clean Room A	P4	LOW	HEPA filter differential pres...	Marcus Miller	6/14/2026	RESOLVED	

Section 5: Artifact/Deliverable

project files:

> This PC > Local Disk (C:) > Asu Homework > EquipmentMaintenanceTracker >				
Name	Date modified	Type	Size	
.vscode	6/14/2026 10:38 PM	File folder		
__pycache__	6/14/2026 11:10 PM	File folder		
static	6/14/2026 10:35 PM	File folder		
templates	6/15/2026 12:36 AM	File folder		
app.py	6/15/2026 1:11 AM	Python Source File	17 KB	
maintenance_records.csv	6/15/2026 1:40 AM	Comma Separate...	3 KB	
README.txt	6/14/2026 11:04 PM	Text Document	3 KB	
requirements.txt	6/14/2026 10:34 PM	Text Document	1 KB	

Cloud Link: hdnguy21.pythonanywhere.com

Dashboard

Log Issue

Reports

Equipment List

Settings

Home / Log Issue

ISSUE LOGGING STATION

Technician portal to log active equipment failures and safety hazards.

Monday, June 15, 2026 at 02:30:44 AM

EQUIPMENT ID

e.g. EQ-832

EQUIPMENT NAME

e.g. Main Intake Valve B

EQUIPMENT TYPE

SEVERITY LEVEL